

Ashkenazi Population Structure Reassessed

Coordinate and Genotype-Level Reference-Panel Sensitivity in the Central Mediterranean

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Abstract

Population-genetic placement is conditional on the reference populations supplied to an analysis. This study tests whether the familiar description of Ashkenazi Jews as intermediate between Europe and the Levant is stable when Italian Jewish, Southern Italian, Sicilian, Maltese, and Aegean comparators are included directly. A literature audit identifies a recurring reference-panel pattern across foundational Y-chromosomal studies and landmark autosomal studies: Northern Italians, Tuscans, Sardinians, and Northern or Central Europeans were repeatedly available, whereas non-Jewish Southern Italians and Sicilians were generally absent. The original quantitative analysis is a linked suite of fifteen Global25 coordinate-space experiments. In the expanded modern coordinate panel, the source labeled Italki is the largest individual fitted source, and the strict Italian Jewish and central Mediterranean block totals 54.6 at distance 0.00819655. Removing only Italki worsens distance by 12.49 percent and reallocates most of the missing coefficient to related Italian and central Mediterranean sources. Removing Italki together with Southern Italian, Sicilian, Maltese, and Aegean comparators worsens distance by 47.18 percent, raises the Lombardy proxy from 9.8 to 38.0, and raises the Levantine-continuity block from 22.8 to 52.0.

To test whether this pattern persists beyond compressed coordinate averages, the study adds an individual-level genotype validation using matched primary and strict-pass AADR-derived panels. The primary panel contains 369 samples, the strict-pass panel contains 247 samples, and both panels cover 18 population groups. Analyses used 184,385 shared post-QC autosomal SNPs for FST and 70,530 matched LD-pruned SNPs for PCA and ADMIXTURE. Across PCA centroid distances, genome-wide Hudson FST, and unsupervised ADMIXTURE, Ashkenazi samples consistently ranked closest to central Mediterranean and eastern Mediterranean comparators. The equal-weight cross-method synthesis ranked Sicilian, Turkish Jewish, Maltese, South Italian, and Central Italian as the five nearest comparators. These results do not erase eastern Mediterranean or Levant-related continuity and do not identify a unique historical source population. They show that a comparator architecture repeated across influential literature can create or exaggerate an apparent Europe-Levant cline when central Mediterranean population space is not represented directly. Formal qpWave and qpAdm modeling remains a separate confirmatory step requiring explicit source and outgroup designs.

Keywords: Ashkenazi Jews; population genetics; reference-panel sensitivity; comparator omission; central Mediterranean; Southern Italy; Italian Jews; Global25; AADR; PCA; FST; ADMIXTURE; qpAdm

1. Introduction

Genome-wide studies consistently identify Ashkenazi Jews as a genetically distinctive, endogamous population with ancestry related to both Europe and the eastern Mediterranean. The central interpretive disagreement is not whether multiple ancestry-related signals exist, but how their geographic scale should be described and how much of the apparent Europe-Levant pattern depends on the reference populations selected for analysis.

The expression Europe-Levant intermediate is often treated as though it were an intrinsic property of the target population. In practice, intermediate placement is defined relative to a declared coordinate system and a chosen reference panel. This is particularly important in the Mediterranean, where Northern Italians, Tuscans, Sardinians, Southern Italians, Sicilians, Maltese, Aegean Greeks, and Italian Jews are related but non-interchangeable comparators.

The comparator problem is visible in both early Y-chromosomal studies and foundational autosomal studies. Many influential panels contained Northern Italians, Tuscans, Sardinians, Basques, French, Russians, or related Northern and Central European populations, but did not include non-Jewish Southern Italians and Sicilians as direct competing comparators. When the central Mediterranean space is missing, north-shifted Italian and Levantine references may bracket the absent region and make a clinal interpretation appear more stable than it is.

The present revision adds a genotype-level validation to the earlier coordinate-space sensitivity design. The goal is not to replace the Global25 experiments with a new final ancestry model, but to test whether the same regional ordering appears when individual-level SNP data are analyzed through PCA, FST, ADMIXTURE, and cross-method rank synthesis.

2. Materials and Methods

2.1 Study design

The study combines two reference-panel sensitivity analyses. The first is a secondary analysis of previously published or publicly distributed population-average coordinates and reported population-genetic results. The original coordinate-space contribution is a linked set of fifteen controlled Global25 experiments designed to test reference-panel sensitivity rather than to estimate a final demographic model.

The second layer is an individual-level genotype validation using matched primary and strict-pass panels derived from the Allen Ancient DNA Resource, AADR v66.p1, June 8, 2026 release. This genotype-level stage tests whether the coordinate-space pattern is reproduced under PCA, genome-wide FST, and ADMIXTURE after explicit sample selection, SNP matching, quality control, LD pruning, replicate alignment, and primary-versus-strict sensitivity checks.

Table 1. Empirical layers in the revised manuscript.

Layer	Data frame	Role in manuscript
Coordinate-space sensitivity	Global25 Experiments 1-15	Matched source restoration/omission, reciprocal Italki tests, regional targets, ancient Levant controls, full-dimensional vector diagnostic
Genotype-level validation	AA DR-derived primary and strict-pass panels	PCA centroid distances, Hudson FST with jackknife uncertainty, ADMIXTURE K=2-12, integrated rank synthesis
Formal modeling boundary	ADMIXTOOLS readiness	qpWave/qpAdm require separate source and right-population designs; no qpAdm result is inferred from descriptive analyses

2.2 Coordinate-space experimental settings

The Global25 experiments used scaled population-average coordinates distributed through the declared source files documented in the accompanying Version 1 data package. Runs were recorded with fixed

calculator settings, retained source and target labels, raw outputs, grouped summaries, screenshots, comparison files, validation scripts, and SHA-256 checksums.

Displayed coefficients are nonnegative, constrained, source-panel-dependent similarity weights. Correlated sources can substitute for one another, and an unprinted or zero coefficient does not demonstrate historical absence. For matched omission experiments, fit deterioration was calculated as $100 \times (d_{\text{reduced}} - d_{\text{full}}) / d_{\text{full}}$, where d is the displayed residual distance.

2.3 Individual-level genotype validation panel

The genotype-level validation used a deliberately constructed modern reference panel centered on Ashkenazi, central Mediterranean, Jewish comparator, Levantine, and broader European/Mediterranean populations. The primary panel retained the broad intended sample set after baseline quality control and contained 369 samples across 18 population groups. The strict-pass panel applied more restrictive sample filtering and contained 247 samples across the same population-label framework.

The Ashkenazi target group contained seven samples in both panels, so genotype-level results are interpreted as reference-panel sensitivity tests rather than as high-precision estimates of Ashkenazi population-wide parameters. Genome-wide F_{ST} analyses used 184,385 shared post-QC autosomal SNPs. PCA and ADMIXTURE analyses used a matched LD-pruned autosomal subset of 70,530 SNPs.

2.4 Genotype-level PCA, F_{ST} , ADMIXTURE, and synthesis

PCA was run on the matched LD-pruned genotype datasets. Population centroids were calculated in PC space, and Ashkenazi-to-comparator centroid distances were summarized across PC1-PC10. Primary-versus-strict stability was evaluated using Procrustes alignment across PC1-PC10.

Pairwise population differentiation was estimated on the shared 184,385 post-QC autosomal SNPs. Hudson F_{ST} was calculated as a ratio of summed finite-sample-corrected numerators to summed denominators. Uncertainty was estimated by delete-one-autosome jackknife across chromosomes 1-22. Ashkenazi-versus-comparator rankings were cross-validated with PLINK Weir-Cockerham weighted F_{ST} .

Unsupervised ADMIXTURE was run independently in each panel for $K=2$ through $K=12$, with five deterministic random-seed replicates per K , ten-fold cross-validation, and eight computational threads. Replicates were aligned before averaging. Because ADMIXTURE components are model-based latent clusters, component labels were not interpreted as literal ancient populations or direct ancestry proportions.

For cross-method synthesis, Ashkenazi comparator ranks were integrated across PCA centroid distance, Hudson F_{ST} , and ADMIXTURE population-mean Q -profile distance. The three K values with the lowest mean cross-validation error in each panel were used for ADMIXTURE rank synthesis. Ranks were normalized within each method and panel, then averaged with cross-method and cross-panel dispersion retained as a stability measure.

3. Results

3.1 Coordinate-space source-panel sensitivity

In the full modern coordinate panel, the source labeled Italki is the largest individual fitted source at 40.8, and the strict Italian Jewish and central Mediterranean block totals 54.6 at distance 0.00819655. Lombardy is 9.8, Levantine continuity is 22.8, and Northern/Eastern Europe is 12.8. These values

establish the baseline response when all declared source categories are allowed to compete; they are not treated as ancestry percentages.

Removing only Italki increases distance to 0.00922012, a deterioration of 12.49 percent. Removing Italki together with Southern Italian, Sicilian, Maltese, and Aegean sources increases distance to 0.01206331, a deterioration of 47.18 percent. In that reduced panel, Lombardy rises from 9.8 to 38.0 and Levantine continuity rises from 22.8 to 52.0.

3.2 Reciprocal, regional, and positive controls

Reciprocal Italki models show that the central Mediterranean direction is present in an Italian Jewish target independently of later Ashkenazi residence north of the Alps. When Ashkenazi sources are excluded and German, Polish, and Russian sources are allowed to compete, the Italki target retains central Mediterranean and Levantine-continuity structure without requiring Northern/Eastern European host sources.

Regional Ashkenazi targets reproduce the same category-level ordering despite subgroup and sample-size differences. Ancient Levant positive controls recover strong local Levantine continuity when appropriate earlier Levantine references are supplied, showing that the framework does not automatically force central Mediterranean sources into every eastern Mediterranean target.

3.3 Individual-level genotype validation

The genotype-level validation reproduced the central panel-sensitivity pattern using individual-level AADR-derived data rather than Global25 population averages. The primary panel contained 369 samples and the strict-pass panel contained 247 samples across 18 population groups.

In PCA, the nearest Ashkenazi population centroids using PC1-PC10 were concentrated among central Mediterranean and eastern Mediterranean comparators. In the primary panel, the five nearest centroids were Maltese, Sicilian, South Italian, Turkish Jewish, and Cypriot. In the strict-pass panel, the five nearest centroids were Maltese, Sicilian, Cypriot, Turkish Jewish, and Greek. The primary-versus-strict Procrustes RMSD across PC1-PC10 was 0.0543532167.

PCA centroid distances to Ashkenazi

Distances use PC1-PC10 in matched LD-pruned datasets; primary and strict-pass panels are plotted together to show sensitivity.

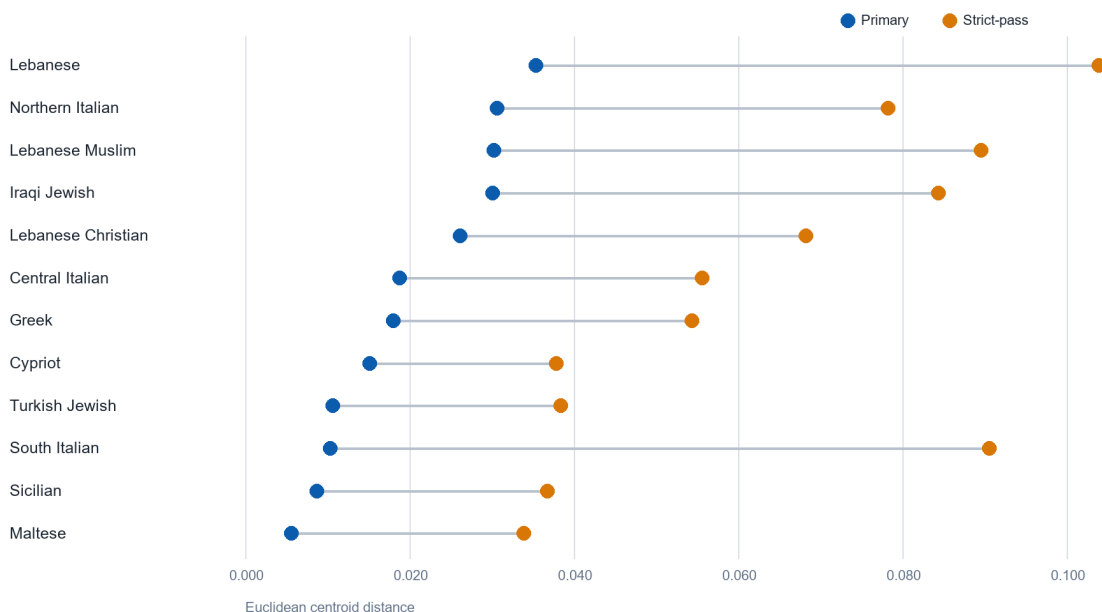


Figure 1. PCA centroid distances to Ashkenazi using PC1-PC10 in the primary and strict-pass LD-pruned datasets. Lower values indicate closer centroid placement within the tested panel.

Genome-wide Hudson FST produced a closely related ordering. In both the primary and strict-pass panels, the five lowest Ashkenazi pairwise Hudson FST comparisons were Turkish Jewish, Sicilian, Central Italian, South Italian, and Greek. Across the 17 Ashkenazi-versus-comparator pairs, primary and strict Hudson FST rankings had Spearman $\rho=0.9093$. Hudson rankings were also concordant with PLINK Weir-Cockerham weighted FST, with Spearman $\rho=0.9583$ in the primary panel and $\rho=0.9093$ in the strict-pass panel.

Ashkenazi pairwise Hudson FST

Lower FST indicates lower allele-frequency differentiation within the tested panel. Top primary-panel comparators are shown with strict-pass values.

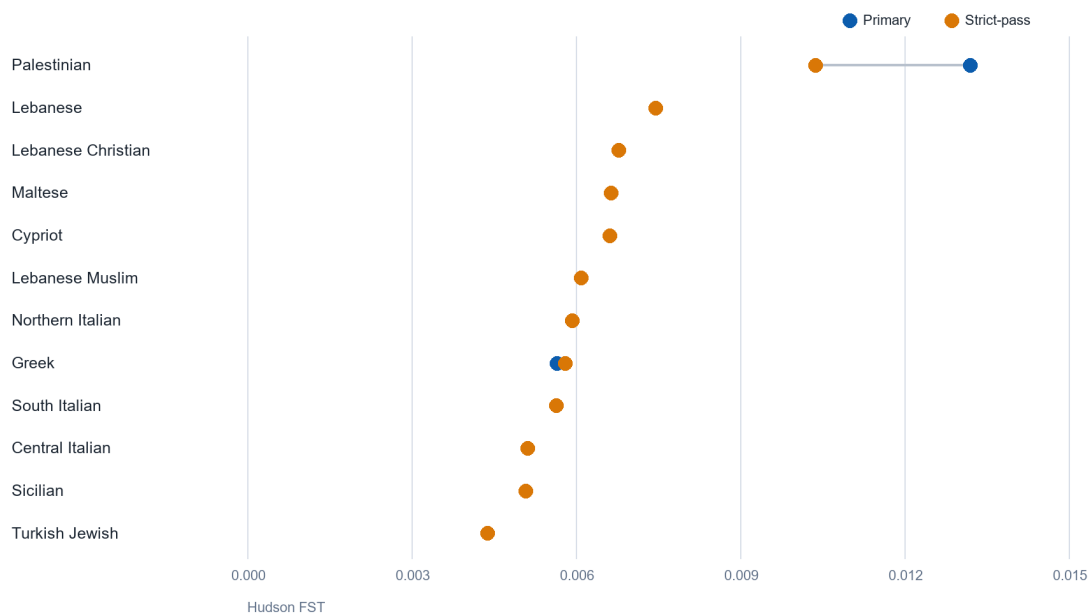


Figure 2. Ashkenazi pairwise Hudson FST for leading comparators. Values are shown for both primary and strict-pass panels; lower FST indicates lower allele-frequency differentiation within the tested panel.

ADMIXTURE also placed Ashkenazi population-mean Q profiles nearest to central Mediterranean and eastern Mediterranean comparators, although the CV-supported K values differed between panels. In the primary panel, the lowest mean cross-validation error occurred at K=12, with K=12, K=11, and K=10 used as the supported K range for rank synthesis. In the strict-pass panel, the lowest mean cross-validation error occurred at K=2, with K=2, K=3, and K=4 used as the supported K range.

ADMIXTURE cross-validation by panel

Primary favors higher K in this panel; strict-pass favors K=2. The manuscript uses supported K ranges rather than a single universal K.

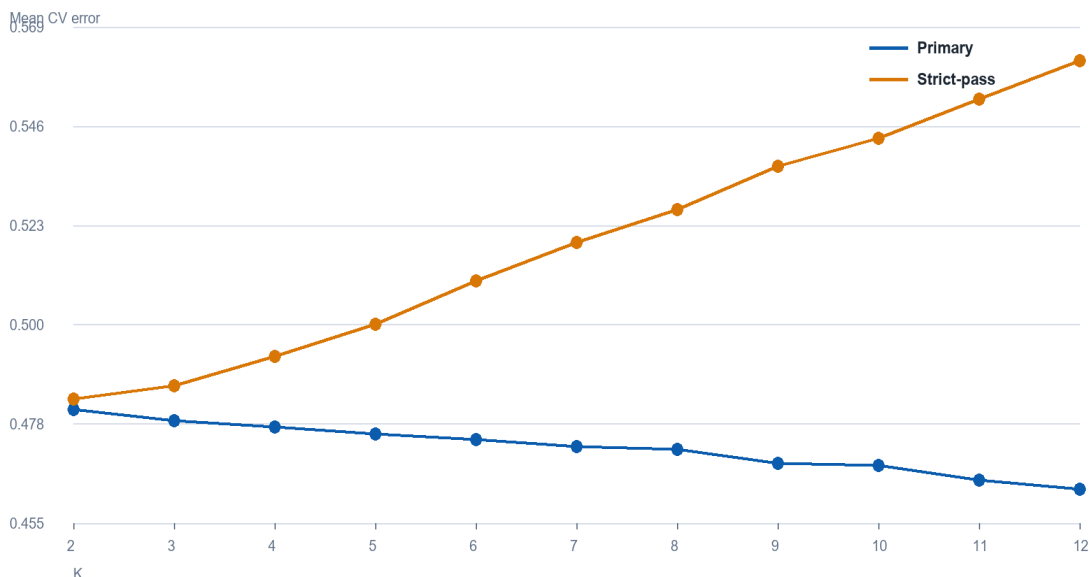


Figure 3. ADMIXTURE cross-validation by panel. The different CV-supported K ranges are treated as a sensitivity result rather than collapsed into one universal K.

ADMIXTURE population-mean Q-profile distances

Primary K=12 and strict-pass K=2 are shown because they are the lowest mean CV K values in their respective panels.

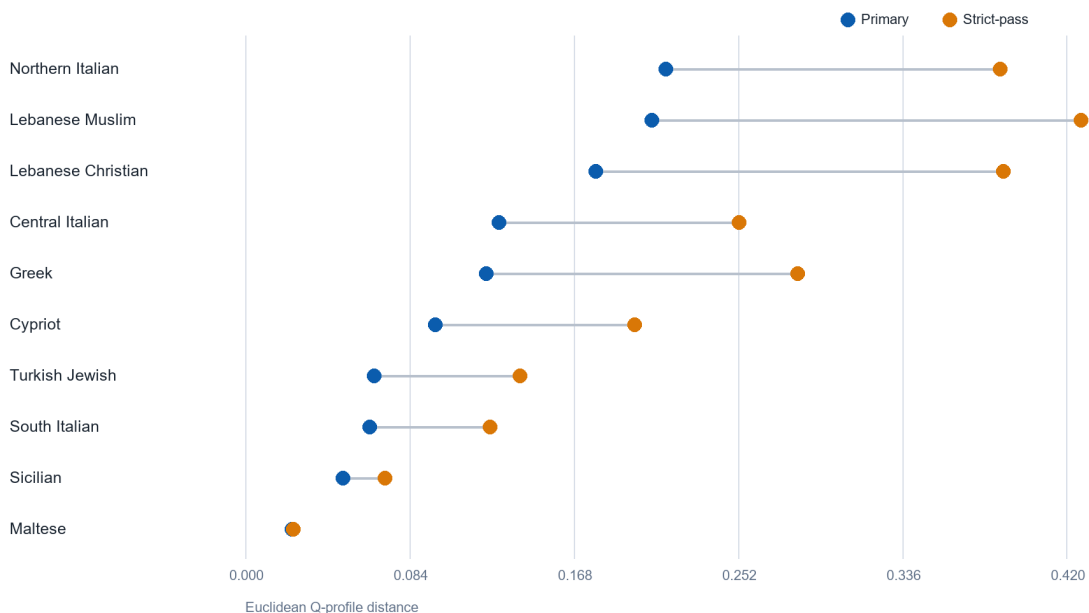


Figure 4. ADMIXTURE population-mean Q-profile distances to Ashkenazi. The displayed comparison uses the lowest mean CV K for each panel: primary K=12 and strict-pass K=2.

3.4 Cross-method genotype synthesis

The equal-weight cross-method synthesis integrated PCA, FST, and ADMIXTURE ranks across the primary and strict-pass panels. The five highest-ranked comparators were Sicilian, Turkish Jewish, Maltese, South Italian, and Central Italian. Greek and Cypriot followed closely. Northern Italian, Tuscan, Bergamo Italian, and Levantine comparators were informative but ranked lower or showed greater panel sensitivity.

Integrated Ashkenazi comparator ranking

Equal-weight synthesis of PCA centroid distance, Hudson FST, and ADMIXTURE profile distance across primary and strict-pass panels.

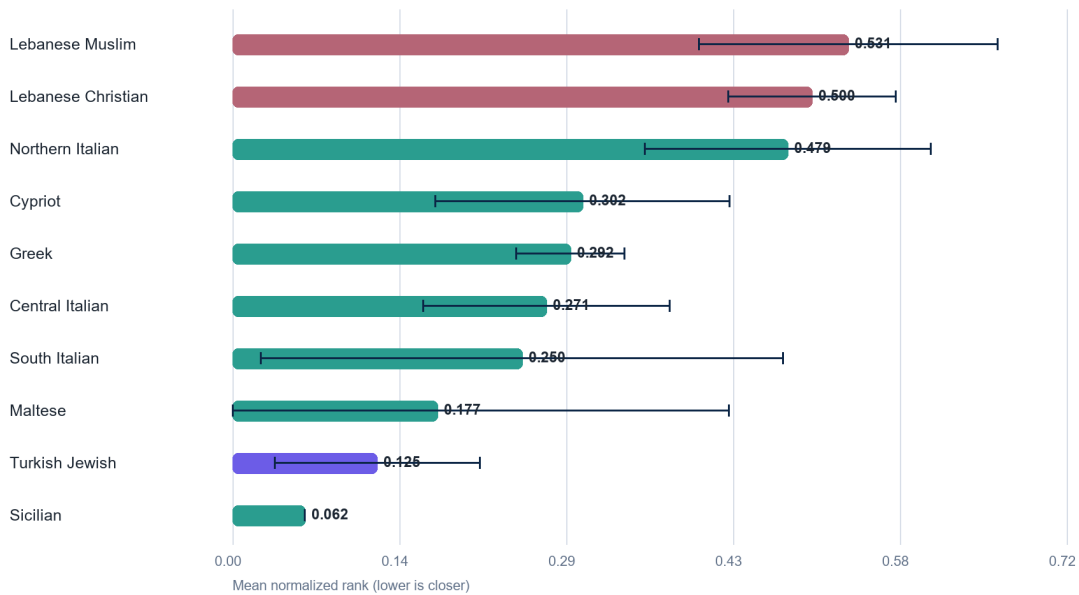


Figure 5. Integrated genotype-level Ashkenazi comparator ranking across PCA centroid distance, Hudson FST, and ADMIXTURE Q-profile distance. Lower normalized rank indicates closer placement within the tested panel; dispersion summarizes cross-method and cross-panel stability.

Table 2. Integrated genotype-level Ashkenazi comparator ranking across PCA, Hudson FST, and ADMIXTURE.

Rank	Comparator	Category	Mean normalized rank	Stability SD
1	Sicilian	Central/east Mediterranean	0.062	0.000
2	Turkish Jewish	Jewish comparator	0.125	0.088
3	Maltese	Central/east Mediterranean	0.177	0.251
4	South Italian	Central/east Mediterranean	0.250	0.225
5	Central Italian	Central/east Mediterranean	0.271	0.106
6	Greek	Central/east Mediterranean	0.292	0.047
7	Cypriot	Central/east Mediterranean	0.302	0.127
8	Northern Italian	Central/east Mediterranean	0.479	0.123
9	Lebanese Christian	Levantine	0.500	0.072
10	Lebanese Muslim	Levantine	0.531	0.129

4. Discussion

The central result is methodological: the apparent Northern Italian-Levantine cline is not stable to the restoration of Italian Jewish and central Mediterranean comparators. In the full coordinate panel, the largest fitted direction is the strict central Mediterranean block. Removing only Italki transfers most of its fitted weight to related Mediterranean sources and modestly worsens distance. Removing the wider central Mediterranean space produces a much stronger Lombardy-Levantine combination but also a 47.18 percent deterioration in distance.

The genotype-level validation strengthens this conclusion because it uses individual-level data and methods independent of the Global25 coordinate-fitting framework. PCA centroid distances, Hudson FST, and ADMIXTURE Q-profile distances do not estimate the same quantity, yet they converge on a similar comparator ordering. Sicilian, Maltese, South Italian, Central Italian, Greek/Cypriot, and Turkish Jewish comparators remain closer than a simple Northern Italian-Levantine framing would predict.

This convergence does not identify one source population. It shows that the central Mediterranean signal is not an artifact of the Global25 coordinate representation alone. It also shows why panel sensitivity should be reported before proxy coefficients, intermediate placement, or ADMIXTURE profiles are translated into historical narratives.

4.1 Boundary between descriptive affinity and formal source modeling

The coordinate-space and genotype-level results both support the same methodological claim: central Mediterranean comparators are necessary for interpreting Ashkenazi placement. They should not be converted directly into a final ancestry model. PCA summarizes covariance, FST measures allele-frequency differentiation, ADMIXTURE estimates latent profiles within a chosen K and panel, and Global25 fitting summarizes coordinate-space approximation. None of these methods alone establishes admixture direction, source chronology, or exact ancestral proportions.

Formal qpWave and qpAdm analyses require a different design. Candidate sources and right populations must be declared in advance, outgroups must be justified, and models must be evaluated by feasibility, p-value, coefficient stability, standard errors, and nested-source competition. Present-day Jewish and Mediterranean populations may be useful comparators, but they should not be treated as literal ancient sources without appropriate ancient-DNA context. The present study therefore uses ADMIXTOOLS readiness as a boundary condition: the descriptive evidence motivates formal testing but does not substitute for it.

5. Limitations

Global25 is a compressed coordinate representation. The coordinate-space experiments cannot audit or repeat upstream marker-level quality control, missingness filters, or LD treatment from the population-average coordinates alone. Their strength is the matched panel perturbation design, not formal parameter estimation.

The genotype-level validation reduces but does not eliminate proxy limitations. The panels are modern comparator panels, and modern populations are imperfect representatives of historical populations. Geographic labels identify sampled variation and do not establish direct descent from the named group.

The Ashkenazi genotype-level target group contains seven samples in both the primary and strict-pass panels. Results should therefore be interpreted as a reference-panel sensitivity test and cross-method

ranking exercise, not as a high-precision estimate of Ashkenazi-wide allele frequencies or source proportions.

ADMIXTURE components are latent clusters and are sensitive to K, sample composition, and panel design. They are used here only as one descriptive profile-distance method integrated with PCA and FST. Raw Q-profile similarity is not interpreted as ancient ancestry proportion.

The study does not yet report new qpWave or qpAdm results. Formal source modeling requires a separately justified ADMIXTOOLS design with declared right populations, ancient-source context where appropriate, model competition, coefficient feasibility filters, standard errors, p-values, and primary-versus-strict replication.

6. Conclusions

The familiar Ashkenazi Europe-Levant cline is not stable under matched changes to the reference panel. With Italki, Southern Italian, Sicilian, Maltese, and Aegean sources present in coordinate-space models, they form the largest fitted direction. When those sources are removed, north-shifted Italian and Levantine proxies absorb the missing signal, and the fit becomes substantially worse.

The individual-level genotype validation supports the same conclusion from an independent analytical layer. Across PCA centroid distances, Hudson FST, and ADMIXTURE Q-profile distances, Ashkenazi samples remain closest to central Mediterranean and eastern Mediterranean comparators, with Sicilian, Turkish Jewish, Maltese, South Italian, and Central Italian ranking highest in the integrated synthesis. These results do not erase eastern Mediterranean or Levant-related continuity. They change the geographic scale and certainty that can be assigned to proxy coefficients and clinal descriptions.

Reference-panel sensitivity should therefore be a standard requirement before intermediate placement, ADMIXTURE profiles, FST rankings, or coordinate-space coefficients are converted into historical ancestry claims. The next decisive stage is formal ADMIXTOOLS testing with predeclared source and outgroup designs, not retrospective interpretation of descriptive proximity metrics as final demographic models.

Data Availability and Reproducibility

The complete Global25 reproducibility repository is available at https://github.com/StevenParker-collab/ashkenazi-population-structure-v1-reproducibility/tree/main/Ashkenazi_Population_Structure_Reassessed_V1_repository.

The genotype-level validation project is archived separately as a complete Stages 01-08 project snapshot containing manifests, sample-selection records, QC records, EIGENSTRAT and PLINK datasets, shared SNP lists, LD-pruned SNP lists, PCA outputs, FST outputs, ADMIXTURE runs, integrated synthesis results, scripts, logs, environment records, inventories, and checksums. Large binary genotype files should be distributed separately from manuscript tables and figures, with raw .bed, .geno, .pgen, .bcf, or VCF files provided only when needed for independent reruns or verification.

Ethics, Funding, and Conflicts

This study reports secondary analyses of previously published or publicly distributed population-level data and introduces no new participant recruitment, biological sampling, genotyping, or sequencing.

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